Distribution-Free Conformal Coverage for IVIM Parameter Maps, and the Identifiability Wall in the Pseudo-Diffusion Compartment

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Abstract

Purpose: To establish and characterize the irreducible identifiability limit that governs per-voxel uncertainty in the ill-posed pseudo-diffusion compartment (D^*) of intravoxel incoherent motion (IVIM) imaging – using distribution-free conformal prediction both as the finite-sample coverage tool that model-based uncertainty quantification (UQ) lacks and as the lens that exposes where *no* label-free method can succeed – and to benchmark that conformal layer head-to-head against model-based IVIM UQ.

Methods: On a controlled, seeded synthetic cohort built from our own bi-exponential forward model with Rician noise, we compare split-conformal and conformalized quantile regression (CQR) against four model-based baselines (probabilistic neural network, mixture-density deep ensemble, deep ensemble, per-voxel Bayesian MCMC), measuring marginal and (SNR $\times$ true-D^*-tercile) conditional coverage and sharpness, with empirical $[5, 95]$ confidence bands on the conditional quantities from a 16-seed re-run. A Fisher-information / Cramér–Rao analysis diagnoses the residual gap; weighted (nonexchangeable) conformal and a label-free deployment monitor stress exchangeability; a qualitative in-vivo demonstration is included.

Results: Model-based IVIM UQ is overconfident *broadly* across D, D^*, and f (not specifically pseudo-diffusion), so the naive marginal hypothesis is rejected. Conformal and conformalized methods restore near-nominal marginal coverage everywhere ($|\text{gap}| \leq 0.024$), and conformalizing the MDN is the sharpest valid recipe (0.65–0.77$\times$ pure-CQR width). The real failure is conditional: the high-D^* regime under-covers for every method (robustly at low-to-moderate SNR), and no label-free method closes the gap because the Cramér–Rao bound on D^* reaches the regime-bin width – an irreducible identifiability limit. The wall is robust to the b-value scheme; weighted conformal recovers covariate shift but not forward-model misspecification; the monitor fires before coverage fails but is blind to the latent gap.

Conclusion: The central finding is an IVIM-specific identifiability boundary: high-D^* regime-conditional coverage is unrecoverable by any label-free method because the Cramér–Rao bound exceeds the regime resolution. Conformal prediction supplies the finite-sample marginal guarantee model-based IVIM UQ lacks – and conformalizing a strong base is the sharpest valid recipe – but its deeper value here is diagnostic: it draws the line the data themselves impose. We characterize that limit; we do not claim to solve it.

Keywords: intravoxel incoherent motion (IVIM); uncertainty quantification; conformal prediction; identifiability; pseudo-diffusion; quantitative MRI.

1 Introduction

IVIM imaging [1] separates microcirculatory perfusion from tissue diffusion by fitting the bi-exponential signal

$$\frac{S(b)}{S_0} = f\, e^{-bD^*} + (1-f)\, e^{-bD}, \tag{1}$$

where b is the diffusion weighting (s/mm^2), D the tissue diffusion coefficient, $D^* \gg D$ the pseudo-diffusion coefficient, and f the perfusion fraction. Equation (1) is notoriously ill-posed: D^* is poorly constrained because the fast-decaying perfusion term contributes only at low b, and small perfusion fractions make it nearly unidentifiable. Clinical use of IVIM parameter maps therefore demands trustworthy *per-voxel uncertainty*, not just point estimates.

The dominant approach to IVIM UQ is *model-based*: probabilistic neural networks, deep ensembles, mixture-density networks, and Bayesian fitting place a distribution over parameters and read intervals from it. These methods make distributional assumptions and carry *no finite-sample coverage guarantee*; the most comprehensive IVIM-UQ study to date [2] documents that such models are well-calibrated for D and f but *overconfident in* D^*. *Conformal prediction* [3–5] offers the complementary property: under exchangeability of calibration and test data, it produces intervals with finite-sample, distribution-free marginal coverage for *any* base predictor – a biased base only widens the interval, it does not break coverage. Conformalized quantile regression (CQR) [6] makes these intervals input-adaptive. Conformal UQ has been applied to quantitative MRI relaxometry and field mapping [7], but never to the ill-posed IVIM inverse problem, and never benchmarked head-to-head against model-based IVIM UQ. We situate both literatures in Sec. 2.

This paper closes that gap and, in doing so, *reframes* the question. We began with the hypothesis that conformal prediction's advantage would concentrate in the unstable D^* (and secondarily f) compartment, exactly where model-based UQ is documented weak. The data reject the *marginal* form of that hypothesis and replace it with a sharper, IVIM-specific finding. Our contributions are:

1. **An irreducible identifiability limit, characterized.** The genuine IVIM-specific phenomenon is conditional: the high-D^* regime under-covers universally, eleven label-free conditioning methods fail to close the gap, and a Fisher-information / Cramér–Rao analysis shows why – the bound on D^* reaches the regime-bin width, so the compartment cannot be localized from the data (Secs. 4.2–4.3). This is the central contribution; the conformal layer below is what renders the limit measurable and proves it is information-theoretic, not methodological.

2. **The first conformal/CQR coverage for IVIM, benchmarked.** We bring distribution-free conformal prediction to the ill-posed IVIM inverse problem with validated finite-sample marginal coverage for (D, D^*, f), and the first head-to-head benchmark against four model-based UQ baselines – showing model-based IVIM UQ is overconfident *broadly* (not D^*-specifically) and that conformalizing a strong model-based base is the sharpest valid recipe (Secs. 3, 4.1).

3. **Robustness, acquisition-sensitivity, and an honest in-vivo demonstration**: weighted conformal recovers covariate shift but not forward-model misspecification, an observable monitor flags deployment-validity breaks before coverage fails, the wall is robust to the b-value scheme, and the in-vivo component is explicitly qualitative with no coverage claim (Secs. 4.4–4.6).

Every number in this paper traces to a gated, seeded checkpoint output; a machine-checked consistency report is included (Sec. 7).

2 Related work

IVIM estimation and its uncertainty. The bi-exponential IVIM fit is long known to be ill-posed in the perfusion compartment, and a substantial literature addresses the resulting instability of D^*: segmented and constrained non-linear least squares, multi-algorithm comparisons that document wide between-method disagreement [**?**], and acquisition design that places b-values to minimize parameter error, including Cramér–Rao-guided schemes [**?**]. A parallel line replaces fitting with learning – supervised and physics-informed deep networks for IVIM estimation [**? ? ?**] – which sharpen point estimates but do not by themselves certify their own reliability. Uncertainty quantification for IVIM is accordingly *model-based*: Bayesian fitting with informative or spatial priors [**? ?**], posterior estimation with explicit uncertainty [**?**], and, most comprehensively, deep ensembles of mixture-density networks that decompose aleatoric and epistemic uncertainty and report calibration on simulated and in-vivo data [2]. Every such method places a distribution over parameters and reads intervals from it; none carries a finite-sample, distribution-free coverage guarantee, and the most thorough of them documents residual overconfidence in D^* [2]. A guarantee that holds regardless of how well the model is specified is exactly what conformal prediction supplies – and is what motivates asking where even a guaranteed method must fail.

Conformal prediction and its imaging applications. Conformal prediction [3–5] yields finite-sample, distribution-free coverage for any base predictor under exchangeability; conformalized quantile regression [6] makes the intervals input-adaptive, and the framework extends to covariate shift [8, 9] and to localized and conditional guarantees [10, 11]. In imaging it has been applied to distribution-free image-to-image regression and uncertainty maps [**?**], to deployment concerns such as fairness in medical-imaging predictors [**?**], and – closest to this work – to quantitative-MRI relaxometry and field mapping via conformalized quantile regression [7]. None of these targets the IVIM inverse problem, none is benchmarked against model-based IVIM UQ, and – the point of this paper – none asks what conformal prediction *reveals* when the data cannot identify the estimand: a finite-sample-valid interval that is honest precisely because it balloons where the parameter is unrecoverable. We turn that honesty into a diagnostic for the IVIM identifiability wall.

3 Methods

Forward model and synthetic cohort. Conformal calibration requires *labeled* data, which in-vivo IVIM cannot provide, so the cohort is synthetic and built from our own implementation of Eq. (1) with Rician noise ($\sigma = S_0/\mathrm{SNR}$). Parameters are drawn uniformly from published abdominal ranges: $D \in [0.5, 3.0] \times 10^{-3}$, $D^* \in [10, 100] \times 10^{-3}\,\mathrm{mm}^2/\mathrm{s}$, $f \in [0.05, 0.40]$; SNR is drawn from $\{10, 20, 30, 50, 100\}$; the b-value scheme has 22 values $(0, 10, \ldots, 100, 120, \ldots, 200, 300, \ldots, 800\,\mathrm{s/mm}^2)$, dense at low b to separate D^*. The three splits (train/calibration/test = 4000/2000/3000) are i.i.d. draws from one seed (`20260613`), making calibration and test exchangeable. On clean (noise-free) signals the generator and the non-linear least-squares (NLLS) estimator recover the truth to a maximum relative error of 2.07×10^{-7}, confirming label self-consistency.

Conformal intervals. For a base point predictor $\hat{\theta}$, *split conformal* uses calibration nonconformity scores $s_i = |\theta_i - \hat{\theta}_i|$ and the radius $q = \lceil (n+1)(1-\alpha) \rceil$-th smallest score, giving $[\hat{\theta} - q,\ \hat{\theta} + q]$ with marginal coverage in $[1-\alpha,\ 1-\alpha+1/(n+1)]$. *CQR* [6] conformalizes a gradient-boosted quantile-regression base. We also *conformalize model-based* bases by treating their predictive quantiles as the CQR band, which restores the guarantee while keeping the model's sharpness.

Table 1: Marginal coverage on test at $\alpha = 0.10$ (nominal 0.90); gap = nominal − realized (positive = overconfident). Model-based methods under-cover broadly; conformal and conformalized variants restore coverage in every compartment. Marginal coverage is finite-sample *guaranteed*; values are the seed-0 (multi-seed index-0) run, with 16-seed MC-precision bands available in the supplementary CI table. From `results/benchmark_report.txt`.

arm	D cov (gap)	D^* cov (gap)	f cov (gap)
raw: PNN-Gaussian	0.787 (+0.113)	0.833 (+0.067)	0.767 (+0.133)
raw: MDN-DeepEnsemble	0.870 (+0.030)	0.876 (+0.024)	0.859 (+0.041)
raw: DeepEnsemble-Point	0.436 (+0.464)	0.425 (+0.475)	0.475 (+0.425)
raw: Bayesian-MCMC	0.876 (+0.024)	0.870 (+0.030)	0.855 (+0.045)
conformal: split-NLLS	0.899 (+0.001)	0.897 (+0.003)	0.907 (−0.007)
conformal: CQR-HGB	0.901 (−0.001)	0.902 (−0.002)	0.893 (+0.007)
conformalized: MDN	0.890 (+0.010)	0.901 (−0.001)	0.882 (+0.018)
conformalized: Bayesian	0.895 (+0.005)	0.894 (+0.006)	0.876 (+0.024)

Model-based baselines. Four standard, independent model-based UQ baselines emit per-parameter intervals on the matched cohort: a single probabilistic NN (Gaussian NLL head); a deep ensemble of mixture-density networks [12, 13] following the Casali approach [2] with an aleatoric/epistemic split; a plain deep ensemble of point regressors (epistemic-only; deliberately weak); and a per-voxel Bayesian fit by component-wise adaptive Metropolis (fed the true noise level, so it is evaluated at its best). A parallel line of work uses normalizing-flow / simulation-based posterior estimation for microstructure inference; Gauge is deliberately *independent* of it (own forward model, own cohort, standard baselines) and does not use it as a baseline.

Conditional coverage and the identifiability diagnostic. We measure coverage on a (true-D^* tercile × SNR) grid – the terciles are a diagnostic only; the true D^* is unknown at test time. To ask whether a residual gap is a method failure or an information limit, we compute the Fisher information for $\theta = (D, D^*, f, S_0)$ under the high-SNR Gaussian approximation, $\mathcal{I} = J^\top J/\sigma^2$ with J the signal Jacobian, and the Cramér–Rao lower bound (CRLB) [14] $\mathrm{CRLB}(\theta_k) = \sqrt{[\mathcal{I}^{-1}]_{kk}}$.

Robustness tooling. To stress exchangeability we use *weighted / nonexchangeable conformal* [8, 9], with importance weights $w(x) = \mathrm{d}P_{\text{test}}/\mathrm{d}P_{\text{cal}}$ estimated by a domain classifier on observable features, and *localized* [10] and *conditional* [11] conformal for the label-free conditioning attack. We also build a label-free *deployment-validity monitor* with two detector families (a Mahalanobis distance on observable signal-summary features and a residual-conformal test on NLLS fit residuals) that fires when an aggregate drift score exceeds a calibration-derived null threshold.

4 Results

4.1 Marginal benchmark: model-based UQ is broadly overconfident

Plain split conformal and CQR track nominal coverage within 0.03 for all parameters and all $\alpha \in \{0.05, 0.10, 0.20, 0.30\}$ on the held-out test split (e.g. split-conformal D^* at α=0.10: realized 0.897 vs nominal 0.90), confirming the conformal layer is correct. Table 1 gives the head-to-head at $\alpha = 0.10$.

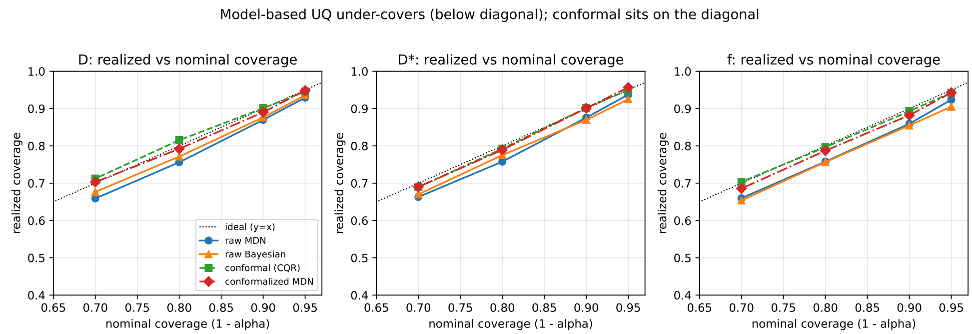

Figure 1: Realized vs nominal coverage per parameter. Raw model-based methods sit below the diagonal (overconfident); conformal and conformalized methods sit on it.

Table 2: High-D^* tercile coverage (D^* parameter, nominal 0.90). Every method under-covers; SNR-Mondrian does not close it. Across-seed means with 16-seed $[5, 95]$ bands. From `results/multiseed_report.txt`.

arm	hi-D^* marginal	hi-D^* worst-SNR cell
raw-MDN	0.841 [0.829, 0.857]	0.753 [0.719, 0.789]
CQR (plain)	0.826 [0.809, 0.851]	0.761 [0.729, 0.795]
split (Mondrian/SNR)	0.829 [0.813, 0.844]	0.781 [0.743, 0.810]
CQR (Mondrian/SNR)	0.829 [0.811, 0.854]	0.789 [0.752, 0.828]
conformalized-MDN	**0.868** [0.856, 0.889]	**0.786** [0.751, 0.826]

Averaging the gap over α, the marginal under-coverage is broad, not D^*/f-concentrated: per-method worst compartments are f (PNN, 0.125), f (MDN, 0.037), D^* (DeepEnsemble-Point, 0.452), and f (Bayesian, 0.045). 0/4 **methods** show D^*/f-concentrated under-coverage; for the MDN ensemble D^* is in fact among the *better*-covered parameters marginally (gap 0.029), because its large aleatoric variance widens its band. The naive marginal hypothesis is therefore *rejected.* Conformal and conformalized variants restore near-nominal coverage everywhere, with maximum conformalized gap 0.024 (Fig. 1). On sharpness (Fig. S1), the cost of restoring coverage is cheap for strong bases (MDN 1.05–1.07×, Bayesian 1.02–1.05× the raw width) and ruinous for the overconfident weak base (DeepEnsemble-Point 3.23–3.76×). Crucially, the model's band helps: *conformalized-MDN is* 0.73/0.77/0.65× *the width of pure CQR* for $D/D^*/f$ at equal guaranteed coverage – the sharpest valid recipe.

4.2 The conditional finding: high-D^* under-covers universally

Marginal coverage hides a compartment-specific failure (Fig. 2). On the (SNR × D^*-tercile) grid, the high-D^* tercile under-covers for *every* method at *every* SNR, while the mid-D^* tercile over-covers (≈ 0.99). Table 2 gives the high-D^* slice: even the best method (conformalized-MDN) reaches only 0.868 [0.856, 0.889] marginal-in-regime and 0.786 [0.751, 0.826] in its worst-SNR cell (16-seed mean and $[5, 95]$ band). Critically, *Mondrian stratification by the known SNR does not fix it* – the failure axis is the *unknown* true D^*, not SNR. This is the refined, IVIM-specific form of the original hypothesis.

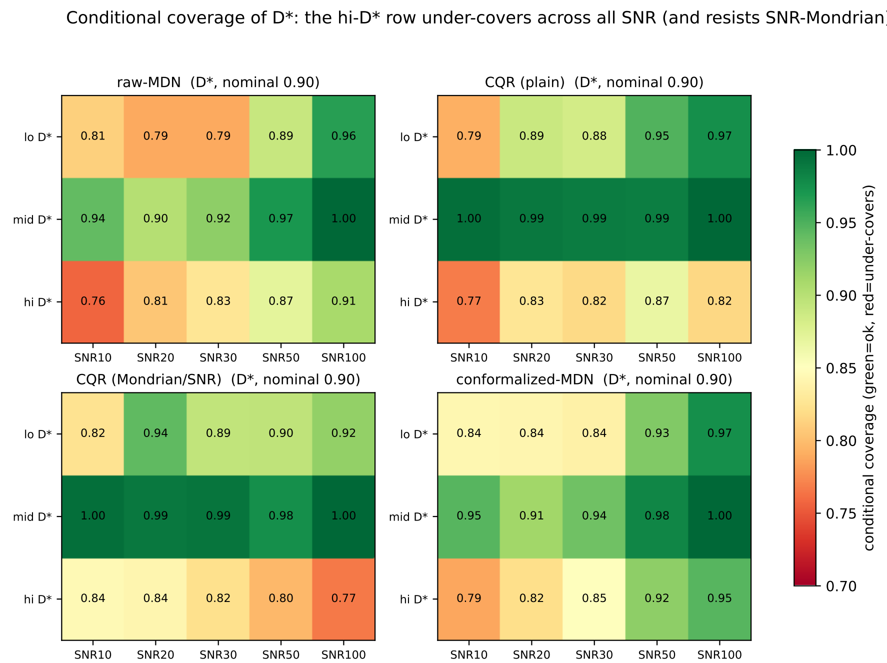

Figure 2: SNR $\times$ D^*-regime conditional coverage. The high-D^* row under-covers robustly at low-to-moderate SNR (recovering toward nominal only at the highest SNR) and resists SNR-Mondrian.

4.3 The high-D^* gap is an irreducible identifiability limit

We attacked the high-D^* gap with eleven *label-free* conditioning methods – plug-in Mondrian by estimated $\hat{D}^*$, localized conformal on observable features [10], and conditional conformal over an observable basis [11] – none of which uses the true D^* (Fig. 3). *None materially closes the gap.* The best label-free method (MDN with localized conformal) reaches $0.874\,{\scriptstyle[0.860,\,0.899]}$ marginal / $0.807\,{\scriptstyle[0.774,\,0.845]}$ worst-SNR, within sampling noise of the 0.786 baseline. The reason is structural, not methodological:

- **Routing error.** A plug-in $\hat{D}^*$ misroutes $\sim 33\%$ ($[30, 35]\%$ over 16 seeds) of true high-D^* voxels to a lower-D^* (smaller-radius) stratum: the estimate is least reliable exactly where it must route correctly. Mondrian-by-$\hat{D}^*$ attains nominal coverage *conditional on the observed stratum* ($[0.90, 0.89, 0.90]$) yet craters on the *true* high-D^* tercile ($0.78\,{\scriptstyle[0.76,\,0.80]}$) – it controls the observable, not the latent axis.

- **The CRLB wall.** The CRLB on D^* grows $\sim 6\times$ from low to high D^* (an analytic, seed-invariant bound); the ratio of median CRLB(D^*) to the tercile bin width rises $0.34 \to 0.67 \to 1.10$ across terciles, the high tercile band $[1.02, 1.15]$ (Fig. 4). In the high-D^* regime the CRLB *exceeds* the bin width: a voxel's D^* cannot be localized to its own tercile from the data, so no observable can route it to a wider-interval stratum. Conformal interval width correctly tracks the per-voxel CRLB (log–log $r = 0.76\,{\scriptstyle[0.72,\,0.77]}$): the intervals are honest, not broken.

Verdict. High-D^* regime-conditional coverage in IVIM is an *irreducible identifiability limit.* We therefore *characterize* the limit; we do not claim to solve it.

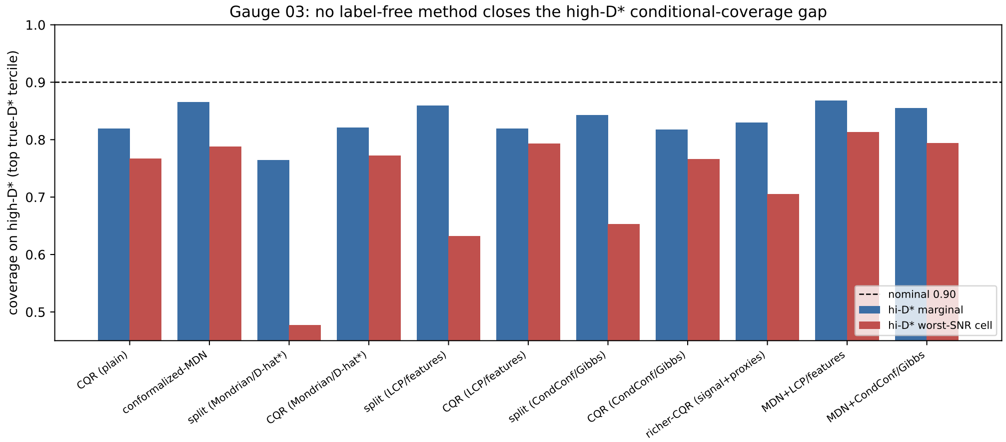

Figure 3: High-D^* coverage (marginal and worst-SNR cell) for eleven label-free conditioning methods; every bar sits below the nominal 0.90.

Table 3: Per-parameter coverage under each break, naive vs. weighted conformal ($\alpha = 0.10$, nominal 0.90), with the deployment monitor's verdict. Across-seed means (16 seeds); per-entry $[5, 95]$ bands are in the supplementary CI table (`results/ci_for_latex.json`). Monitor AUCs are the deterministic seed-0 values. From `results/multiseed_report.txt`.

scenario	method	D	D^*	f	monitor
in-dist (control)	naive	0.898	0.898	0.901	silent (AUC 0.50)
SNR shift (low)	naive	**0.592**	0.739	**0.614**	FIRES (AUC 0.91)
	weighted	0.882	0.901	0.909	
prior shift (harder)	naive	0.972	**0.508**	0.909	FIRES (AUC 0.71)
	weighted	0.919	0.991	0.902	
tri-exp misspec	naive	0.907	0.843	0.912	FIRES (AUC 0.51)
	weighted	0.891	0.933	0.898	
noise-model swap	naive	0.912	0.893	0.906	FIRES (AUC 0.52)

4.4 Robustness under deliberate exchangeability breaks

We deploy one conformal predictor and one monitor, calibrated on the i.i.d. Rician cohort, against four deliberate breaks (Table 3, Fig. S2). Every break miscalibrates coverage; the dangerous undercoverage reaches $D^* = 0.508\,[0.479, 0.543]$ under a harder-tissue prior shift and the worst parameter $0.592\,[0.557, 0.637]$ under a low-SNR covariate shift. *Weighted / nonexchangeable conformal* [8, 9] recovers the covariate shifts (SNR worst-parameter $0.592 \to 0.882$, all within 0.018 of nominal; prior D^* $0.508 \to 0.991\,[0.974, 1.000]$) and is honestly limited on the tri-exponential forward-model misspecification: a shift in the conditional law $P(y \mid x)$ is, in general, not repairable by an x-space likelihood ratio (weighting over-corrects D^* to $0.933\,[0.924, 0.941]$, leaving a per-parameter spread of 0.033).

The label-free monitor (reusing the deployment-validity concept of a sibling project) **fires on all four observable breaks before coverage fails**: in an SNR-severity sweep, it raises the alarm at SNR 20 (coverage still 0.866) while coverage first crosses the failure line at SNR 13. Yet it is constitutively *blind* to the latent high-D^* gap: on an in-distribution (exchangeable) test the monitor is silent (AUC 0.50) and marginal D^* coverage is nominal ($0.898\,[0.880, 0.913]$), while the high-

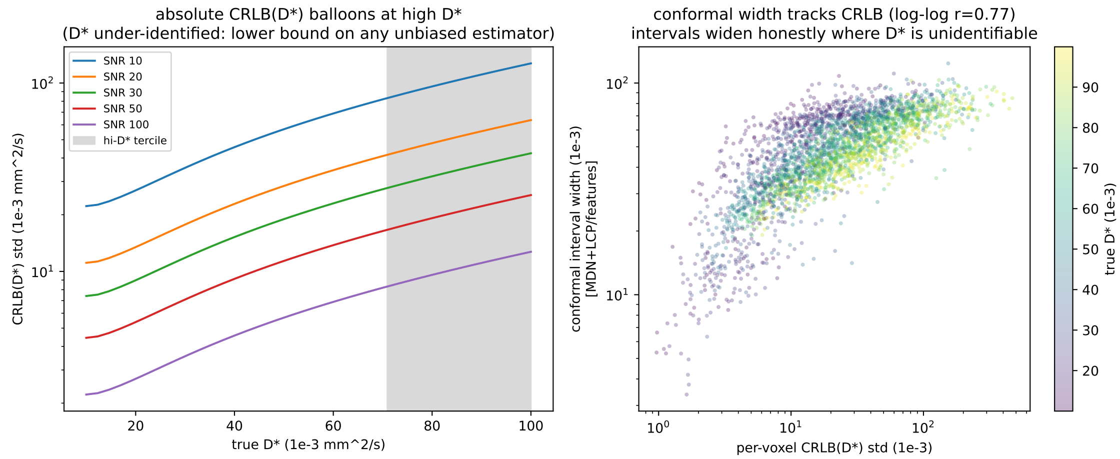

Figure 4: (Left) absolute CRLB(D^*) balloons across the D^* range per SNR (high-D^* tercile shaded). (Right) conformal width vs per-voxel CRLB ($r = 0.76$): the intervals widen honestly where D^* is under-identified.

Table 4: High-D^* coverage and resolution ratio by b-value scheme. Across-seed means with 16-seed $[5, 95]$ bands; the CRLB-optimal ratio band crosses unity. From `results/multiseed_report.txt`.

scheme	n_b	hi-D^* marg	hi-D^* worst-SNR	CRLB/tercile-width
clinical	11	0.821 [0.794, 0.845]	0.596 [0.542, 0.702]	1.24 [1.17, 1.37]
CRLB-optimal	11	0.835 [0.795, 0.862]	0.603 [0.541, 0.678]	**1.03** [0.97, 1.16]
dense	22	0.808 [0.788, 0.836]	0.551 [0.508, 0.597]	1.11 [1.05, 1.21]

D^* conditional coverage is 0.814 [0.787, 0.839]. Observable shifts are caught; the within-distribution latent-axis failure is not – the same observable-vs-hidden split that recurs across this line of work.

4.5 Acquisition sensitivity: the wall is b-scheme-robust

Does the high-D^* wall move with the b-value scheme? We compare a clinical (11-value), a CRLB-optimal (11-value, greedy D^*-CRLB design), and a dense (22-value) scheme at fixed prior/SNR/seed (Table 4, Fig. 5). The high-D^* marginal coverage barely moves ($0.821 \to 0.835 \to 0.808$). The resolution ratio CRLB(D^*)/tercile-width stays at or above unity: even a CRLB-optimal design only reaches ∼unity (1.03 [0.97, 1.16]); acquisition shifts the wall to threshold, not past it, while the clinical and dense schemes stay above it (1.24, 1.11). A CRLB-optimal design lowers the ratio ($1.24 \to 1.03$) but does not remove the wall. Acquisition design *shifts* the wall; it does not erase it – a concrete, honestly negative handoff to acquisition-aware experiment design.

4.6 In-vivo: a qualitative demonstration, no coverage claim

In vivo there are no ground-truth parameters, so realized coverage is *unmeasurable* and the finite-sample guarantee is *not asserted*. We demonstrate the deployed pipeline qualitatively (Fig. S3): the conformalized D^* band widths are wide and heavy-tailed (10/50/90th percentile = $47/64/79 \times 10^{-3}$ mm^2/s; widest-to-narrowest decile ratio 1.7×), ballooning on voxels where the perfusion compartment is under-identified – the wall, now on signal. Pointed at the deployment data with a synthetic calibration, the deployment monitor **fires** (AUC 0.84): synthetic→deployment is itself a

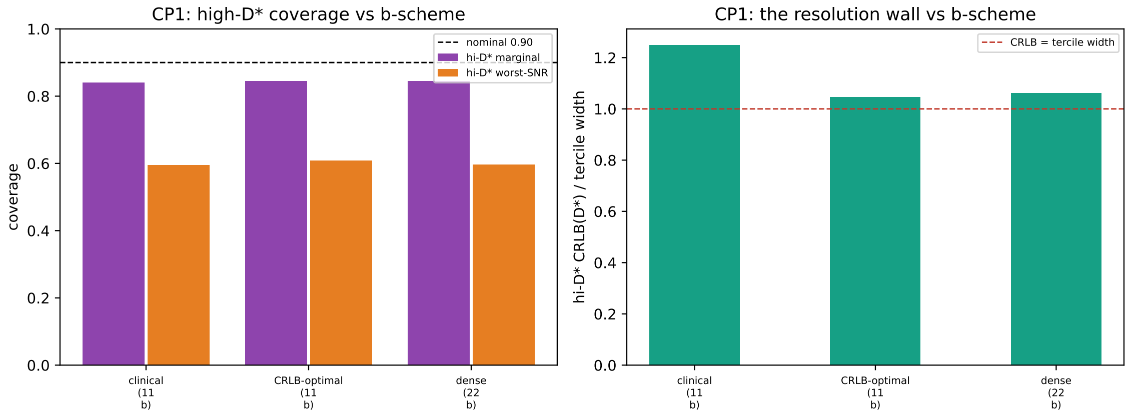

Figure 5: (Left) high-D^* coverage vs. b-scheme. (Right) the resolution ratio CRLB(D^*)/tercile-width stays at or above unity for every scheme (the CRLB-optimal band dips to 0.97): acquisition moves the regime to the resolution threshold, not past it.

large exchangeability break, and the monitor detects it. That firing is the honest, label-free reason the synthetic guarantee must not be transferred in vivo without re-calibration on matched labeled data – which in vivo does not exist. The monitor turns "no ground truth" into an observable stop sign. The demonstration above uses a clearly labeled synthetic stand-in; Sec. 4.7 repeats the *as-deployed* pipeline on a real, permissively-licensed in-vivo collection, where the monitor fires on the synthetic→real transfer for the same reason.

4.7 Real in-vivo data: the same pipeline, the monitor fires on transfer

We additionally run the as-deployed pipeline on real multi-b in-vivo DWI from the public ACRIN-6698 / I-SPY2 Breast DWI collection (The Cancer Imaging Archive; CC-BY-4.0) [15, 16], retrieved on demand (no pixel data are redistributed with this work). This is a 4-b ADC protocol ($b = \{0, 100, 600, 800\}\ \mathrm{s/mm^2}$), *not* IVIM-optimized, with no ground-truth IVIM parameters; it therefore supports only qualitative pipeline/monitor behavior and makes **no in-vivo coverage claim**. A predictor trained on the synthetic 22-value scheme cannot ingest a 4-b signal, and interpolation would fabricate unacquired b-values, so we keep the deployment *monitor* as-deployed—its observable features are b-independent—and re-fit only the CQR band at the real 4-b scheme; the resulting bands are qualitative, not coverage intervals.

Figure 6 shows the actual images on a representative slice: the real b=0 breast DWI, the plug-in $\hat{D}^*$ map, and the per-voxel conformal D^* band-width (uncertainty) map, with the whole-tumor ROI outlined. The uncertainty map is spatially heterogeneous and widens exactly where the perfusion compartment is under-identified—the identifiability wall, now visible on real tissue.

On 4000 foreground voxels the as-deployed monitor **fires** (AUC 0.97) and the D^* band stays wide (10/50/90th percentile = $64/79/86 \times 10^{-3}\ \mathrm{mm^2/s}$; Fig. 7). The firing is carried entirely by the b-independent Mahalanobis family (Family-1); the residual family (Family-2) is silent because a 4-b acquisition exactly-determines the four-parameter IVIM fit, collapsing the fit-residual norm to zero. The monitor has therefore *not* half-failed—it has localized the break to the feature distribution. The sparse-protocol-plus-real-tissue gap is a large exchangeability break, which is exactly why the synthetic guarantee *correctly refuses* to transfer.

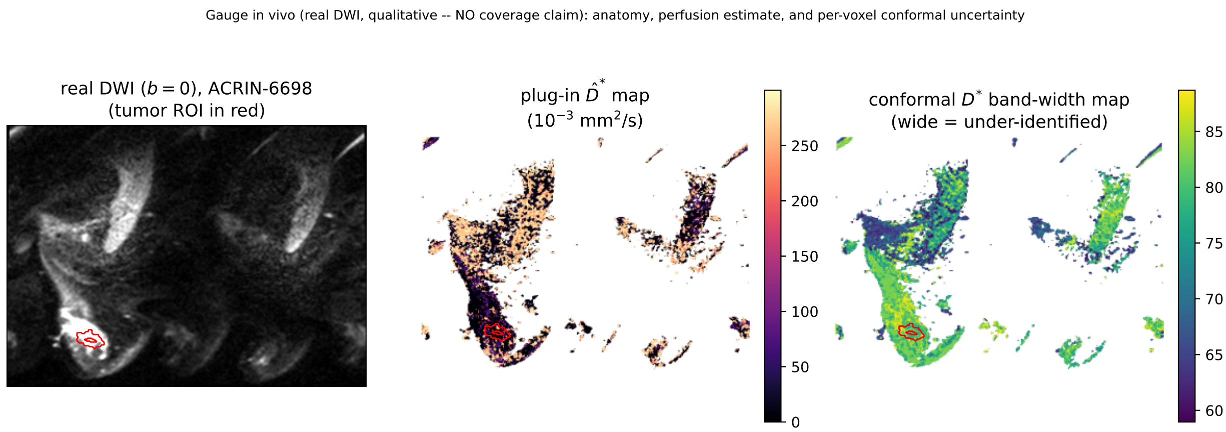

Figure 6: Actual in-vivo images (ACRIN-6698 / I-SPY2 Breast DWI; one slice; qualitative, no coverage claim). (Left) real b=0 DWI with the whole-tumor ROI in red; (middle) the plug-in $\hat{D}^*$ perfusion map; (right) the per-voxel conformal D^* band-width map—wider where the perfusion compartment is under-identified. The synthetic 22-value-calibrated monitor (not shown here) fires on this data; see Fig. 7.

Test–retest repeatability. ACRIN-6698 includes a same-day test–retest arm: two coffee-break repeat DWI exams acquired in a single visit (tagged `TrT0`/`TrT1`), which we use here. (This genuine same-visit arm supersedes an earlier n=11 pairing that had inadvertently matched *longitudinal* timepoints months apart, confounded by interval treatment.) Across n=76 tumors we ask, label-free, whether the conformal interval width *tracks* the scan–rescan variability of the plug-in estimate (region-level over the whole-tumor ROI; the two repeats are unregistered, so this is not a per-voxel statement). For the well-identified tissue diffusion D, the conformal D-width is associated with the measured scan–rescan $|\Delta\text{ADC}|$ (Spearman $r = +0.60$, $p < 0.001$, 95% CI $[0.42, 0.72]$ by BCa bootstrap): the interval *widens where the real measurement is least repeatable.* At n=76 the association is now **robust** (the CI excludes zero, versus the wide interval at the earlier n=11), and stable across field strength (1.5 T r=+0.58, n=49; 3.0 T r=+0.50, n=27) and pretreatment-only (r=+0.61, n=65). For D^* the association is not significant ($r = -0.17$, $p = 0.13$, 95% CI $[-0.39, 0.05]$ spanning zero), consistent with the identifiability wall—the relationship is demonstrable for the parameter the data identify and unresolvable for the one they do not. This is a repeatability-tracking check, *not* a statement of validation, accuracy, calibration, or coverage.

4.8 External-phantom replication on the OSIPI IVIM reference

Every coverage number above is, by necessity, computed on *our* labeled synthetic cohort – our forward model and our parameter prior. To neutralise the "validated only on the authors' own forward model" concern on the part that matters, we replicate the coverage results and the high-D^* wall against an **external, community-standard *synthetic* reference**: the OSIPI (ISMRM Open Science Initiative for Perfusion Imaging) TF2.4 IVIM digital reference object (DRO), 5000 voxels with ground-truth (D, D^*, f) that *we did not generate* [17] (Zenodo `doi:10.5281/zenodo.14605039`, CC-BY-4.0; fetched on demand, provenance only committed). This is an *independent synthetic* reference, **not** in vivo: in-vivo coverage validation remains structurally impossible (no ground truth), so no in-vivo coverage claim is made here. The bi-exponential signal model is the field's consensus model and is therefore shared by construction; what is external is the ground-truth *parameter*

regime	calibration	D	D^*	f	hi-D^* tercile
controlled	naive transfer	0.597 [0.567, 0.621]	0.304 [0.285, 0.315]	0.495 [0.472, 0.512]	0.000 [0.000, 0.000]
controlled	recalibrated	0.899 [0.881, 0.913]	0.905 [0.889, 0.921]	0.904 [0.892, 0.919]	0.823 [0.786, 0.853]
native	naive transfer	0.693 [0.643, 0.758]	0.291 [0.263, 0.312]	0.524 [0.497, 0.542]	0.000 [0.000, 0.000]
native	recalibrated	0.898 [0.888, 0.913]	0.899 [0.884, 0.909]	0.894 [0.882, 0.908]	0.831 [0.798, 0.869]

Table 5: Conformal/CQR coverage on the OSIPI external *synthetic* DRO at nominal 0.90 (CQR; across-seed mean over 16 seeds, $[5, 95]$ band). *Naive transfer* of our synthetic calibration breaks under the real D^* distribution shift; *recalibrating* within the OSIPI distribution restores marginal coverage to nominal on independent ground truth, yet the high-D^* conditional coverage stays below nominal – the identifiability wall, on data we did not generate.

distribution. That distribution is genuinely shifted from ours: the OSIPI true D^* spans ~ 0.05–$0.20\,\mathrm{mm}^2/\mathrm{s}$, above our calibration prior (0.01–0.10), so transferring our calibration onto it is a real out-of-distribution test. All numbers are across-seed means with $[5, 95]$ bands over 16 seeds; nominal coverage is 0.90.

We evaluate two regimes – *controlled* (OSIPI ground truth, re-synthesised at *our* 22-b acquisition and SNR grid, so only the parameter distribution is swapped) and *native* (the DRO's own pre-noised signals at its native sparse 7-b scheme) – each under *naive transfer* (calibrate on our synthetic cohort, test on OSIPI) and *recalibrated* (split the OSIPI DRO and calibrate within its distribution). Table 5 gives the marginal coverage and the top-D^*-tercile conditional coverage; Fig. 8 plots them with the wall metric.

Naive transfer breaks, and the monitor says so. Transferring our synthetic calibration onto the OSIPI ground truth breaks marginal coverage (controlled D^* 0.304 [0.285, 0.315], D 0.597 [0.567, 0.621]), exactly as the exchangeability theory predicts for a real distribution shift. Critically, the label-free deployment monitor *fires on every one of the* 16 *seeds* (Mahalanobis AUC 0.737 [0.726, 0.750], controlled): the shift is observable, so the monitor correctly refuses the transfer. As in the sparse-protocol in-vivo case (§4.7), a fired monitor here is the system working, not failing.

Recalibrated coverage holds on independent ground truth. Splitting the OSIPI DRO and recalibrating within its own distribution restores marginal coverage to nominal for all three parameters (D 0.899 [0.881, 0.913], D^* 0.905 [0.889, 0.921], f 0.904 [0.892, 0.919]; native nearly identical) – the finite-sample conformal guarantee transfers cleanly to a forward model and parameter distribution we did not author.

The high-D^* wall replicates – it is information, not our model. Even after within-distribution recalibration, coverage conditional on the true D^* tercile craters in the high-D^* regime (0.823 [0.786, 0.853] controlled, 0.831 [0.798, 0.869] native) while the low-D^* tercile sits at nominal (0.899 [0.859, 0.931]). On the OSIPI ground truth at our acquisition the CRLB(D^*)/tercile-width ratio rises $0.46 \to 0.86 \to 1.53$ ([1.407, 1.753] at high D^*) with the absolute CRLB growing $3.5\times$ [3.1, 4.0] – the same wall as Fig. 4, on a digital reference object we did not build. (In the native high-SNR regime the CRLB ratio stays below 1, yet the conditional gap persists, consistent with the observable-vs-latent reading: the gap is about conditioning on an axis the data do not identify, not raw CRLB magnitude.) That the wall reproduces on an external ground truth is strong evidence it is information-theoretic, intrinsic to the bi-exponential model and the data, rather than an artifact of our particular forward model or prior.

5 Discussion

The thread connecting these results is an *observable-vs-latent* distinction. Conformal prediction guarantees *marginal* coverage and, with weighting or stratification on *observable* quantities (SNR, signal features, estimated density ratios), can recover coverage conditional on those observables. What it cannot label-free-guarantee is coverage conditional on a *latent* axis that the data do not identify. In IVIM that latent axis is the true D^* in its ill-posed regime: the Fisher information there collapses, the CRLB exceeds the bin width, and no observable proxy – including the plug-in $\hat{D}^*$ – recovers the regime. This is why an observable deployment monitor catches every *distribution* shift but is blind to the *within-distribution* high-D^* gap.

The same wall appears in adjacent label-free problems: deployment-validity monitoring is provably blind to shifts that leave observable statistics unchanged (a "hidden channel"), and the absence of in-vivo ground truth (the "Echo tension") forecloses any direct coverage check on real data. We read these not as failures of conformal prediction – whose guarantee is exactly as advertised – but as a recurring *identifiability* boundary: distribution-free tools deliver everything the data identify and nothing they do not. Practically, this argues for (i) conformalizing a strong model-based base (sharpest valid intervals), (ii) deploying an observable monitor to guard exchangeability, and (iii) reporting the high-D^* compartment as characterized-but-unresolved rather than trusting a point map there.

6 Limitations

Our coverage results are *synthetic-primary* by necessity: conformal calibration needs labels that in-vivo IVIM lacks, so the guarantee is established on a forward model we control, and the in-vivo component is explicitly qualitative with no coverage claim. We stress that this is a structural impossibility, not a study limitation removable with more data: because in-vivo IVIM admits no ground-truth (D, D^*, f), realized coverage is unmeasurable in vivo in principle, so a distribution-free coverage guarantee can only ever be synthetic-established and then transported under an explicitly monitored exchangeability assumption – precisely what the real-data section (Sec. 4.7) exercises, rather than a coverage claim it could never support. The synthetic cohort uses uniform priors over published ranges and Rician noise; real tissue is more heterogeneous, and the tri-exponential and prior-shift stresses (Sec. 4.4) are deliberately schematic probes of misspecification rather than a tissue model. The CRLB diagnostic uses the high-SNR Gaussian approximation to Rician noise; the qualitative under-identification of D^* is noise-model-independent (the D^* Jacobian column vanishes as D^* grows), but absolute bound values at SNR $\lesssim 10$ would shift under an exact Rician treatment. Finally, all coverage guarantees assume exchangeability of calibration and deployment data; the monitor quantifies, but cannot by itself repair, departures from it.

7 Consistency check (reproducibility)

Every numeric claim in this manuscript is drawn from a gated, seeded checkpoint printout. The script `gauge/paper/consistency.py` enforces two contracts: (i) every finite-sample-guaranteed marginal value and every determinism-gated quantity appears *verbatim* in its seed-`20260613` report (`results/*.txt`, `POSITIONING.md`); and (ii) every conditional / empirical (E) headline ships as the *across-seed mean* over a 16-seed re-run and is asserted to lie within its empirical $[5, 95]$ band, which carries an `n_seeds` (`results/multiseed.json`, `multiseed_report.txt`; old→new point map in `results/ci_rebaseline_table.txt`). The full pipeline regenerates deterministically from seed 20260613 (`python -m gauge.baselines`; `gauge.benchmark`; `gauge.conditional`; `gauge.conditional_attack`; `gauge.robustness`; `gauge.invivo`), and the bands from the multi-seed driver (`python -m gauge.multiseed`); seed-0 outputs are byte-identical across independent full re-runs.

8 Conclusion

Gauge brings distribution-free conformal coverage to the ill-posed IVIM inverse problem and benchmarks it against model-based UQ. The naive hypothesis – that the conformal advantage concentrates marginally in D^*/f – is rejected: model-based IVIM UQ is overconfident broadly, conformal restores coverage everywhere, and conformalizing the MDN is the sharpest valid recipe. The genuine IVIM-specific phenomenon is conditional: the high-D^* regime under-covers universally, and a Fisher-information analysis shows this is an irreducible identifiability limit that no label-free method can close. We therefore characterize the limit – and provide the robustness tooling, acquisition analysis, and qualitative in-vivo demonstration that frame how an honest, distribution-free IVIM uncertainty map should be deployed and monitored.

Data and Code Availability Statement

All coverage results are produced from a deterministic, seeded synthetic cohort (seed 20260613) generated by the study's own forward model; no external or in-vivo dataset is required to reproduce them. The complete source code, the gated checkpoint outputs from which every number in this paper is drawn, the figures, and a machine-checked consistency script (`gauge/paper/consistency.py`) are openly available in the project repository and permanently archived at https://doi.org/10.5281/zenodo.20686274. The qualitative in-vivo cross-check (Sec. 4.7) applies the as-deployed predictor and monitor to real multi-b DWI from the publicly available ACRIN-6698 / I-SPY2 Breast DWI collection (The Cancer Imaging Archive; DOI `10.7937/tcia.kk02-6d95`; licensed CC-BY-4.0) [15, 16], which has no ground-truth IVIM parameters and supports only qualitative pipeline/monitor behavior and a label-free same-day test–retest repeatability-tracking check (n=76 tumors, Sec. 4.7)—it makes no in-vivo coverage claim. Under a download-on-demand posture no pixel data are redistributed here; the data are retrieved by `scripts/fetch_invivo.py` into an ignored path, and only the provenance manifest (`results/invivo_real_provenance.json`, with series identifiers, b-values, license, and citation) is committed. The ACRIN-6698 imaging is publicly available and de-identified, requiring no additional ethical approval for this secondary, qualitative analysis; the synthetic coverage pipeline uses no human or animal data and regenerates independently of any external dataset. The external-phantom replication (Sec. 4.8) uses the OSIPI TF2.4 IVIM reference digital reference object—an independent *synthetic* ground truth (code Apache-2.0; data Zenodo `doi:10.5281/zenodo.14605039`, CC-BY-4.0) [17], retrieved on demand by `scripts/fetch_osipi.py` with only the provenance manifest (`results/osipi_provenance.json`, with record DOI, checksums, units, and citation) committed; it is a synthetic reference, not in vivo, and makes no in-vivo coverage claim.

Conflict of Interest

The author declares no competing interests.

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Author Contributions

Avery Karlin is the sole author and conceived the study, designed and implemented the methods and software, performed the analysis, produced the figures, and wrote the manuscript.

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References

[1] Denis Le Bihan, Eric Breton, Denis Lallemand, M-L Aubin, J Vignaud, and M Laval-Jeantet. Separation of diffusion and perfusion in intravoxel incoherent motion MR imaging. *Radiology*, 168(2):497–505, 1988.

[2] et al. Casali. A comprehensive framework for uncertainty quantification of voxel-wise supervised models in IVIM MRI. *NMR in Biomedicine*, 2026. arXiv:2508.04588.

[3] Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, 2005.

[4] Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.

[5] Anastasios N Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

[6] Yaniv Romano, Evan Patterson, and Emmanuel J Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1905.03222.

[7] Felix Birk, Lucas Mahler, Julius Steiglechner, Qi Wang, Klaus Scheffler, and Rahel Heule. Distribution-free uncertainty estimation in multi-parametric quantitative MRI through conformalized quantile regression. Proc. Intl. Soc. Mag. Reson. Med. (ISMRM), abstract #2228, 2024. `https://archive.ismrm.org/2024/2228.html`.

[8] Ryan J Tibshirani, Rina Foygel Barber, Emmanuel J Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. arXiv:1904.06019.

[9] Rina Foygel Barber, Emmanuel J Candès, Aaditya Ramdas, and Ryan J Tibshirani. Conformal prediction beyond exchangeability. *The Annals of Statistics*, 51(2):816–845, 2023.

[10] Leying Guan. Localized conformal prediction: a generalized inference framework for conformal prediction. *Biometrika*, 110(1):33–50, 2023.

[11] Isaac Gibbs, John J Cherian, and Emmanuel J Candès. Conformal prediction with conditional guarantees. *arXiv preprint arXiv:2305.12616*, 2023.

[12] Christopher M Bishop. Mixture density networks. *Technical Report NCRG/94/004, Aston University*, 1994.

[13] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[14] Harald Cramér. Mathematical methods of statistics. *Princeton University Press*, 1946.

[15] David C Newitt, Savannah C Partridge, Zheng Zhang, Jessica Gibbs, Thomas Chenevert, Mark Rosen, Patrick Bolan, Helga Marques, Justin Romanoff, Lisa Cimino, Bonnie N Joe, Heidi Umphrey, Haydee Ojeda-Fournier, Basak Dogan, Karen Y Oh, Hiroyuki Abe, Jennifer Drukteinis, Laura J Esserman, and Nola M Hylton. ACRIN 6698/I-SPY2 breast DWI. The Cancer Imaging Archive, 2021. Data set, license CC-BY-4.0. https://doi.org/10.7937/tcia.kk02-6d95.

[16] Kenneth Clark, Bruce Vendt, Kirk Smith, John Freymann, Justin Kirby, Paul Koppel, Stephen Moore, Stanley Phillips, David Maffitt, Michael Pringle, Lawrence Tarbox, and Fred Prior. The cancer imaging archive (TCIA): maintaining and operating a public information repository. *Journal of Digital Imaging*, 26(6):1045–1057, 2013.

[17] Oscar Jalnefjord, Ivan A Rashid, Daan Kuppens, Merel M van der Thiel, Petra J van Houdt, Paulien H M Voorter, Eric T Peterson, and Oliver J Gurney-Champion. OSIPI TF2.4 IVIM-MRI code collection and reference phantoms. ISMRM Open Science Initiative for Perfusion Imaging (OSIPI); software v0.1.0, Apache-2.0, https://github.com/OSIPI/TF2.4_IVIM-MRI_CodeCollection, 2026. Reference-phantom data set: Zenodo, license CC-BY-4.0, https://doi.org/10.5281/zenodo.14605039. Accompanying paper in Magnetic Resonance in Medicine (in press).

Figure 7: Real in-vivo data (ACRIN-6698 / I-SPY2 Breast DWI; qualitative, no coverage claim). (Top left) the re-fit conformal D^* band vs the plug-in $\hat{D}^*$; (top right) the D^* band-width distribution and the as-deployed monitor verdict (fires, AUC 0.97, via the b-independent Mahalanobis family). (Bottom) test–retest: the conformal D-width vs the measured scan–rescan ADC variability across n=76 same-day-repeat tumors (Spearman r = +0.60, 95% CI $[0.42, 0.72]$; robust, repeatability-tracking only—not validation or coverage).

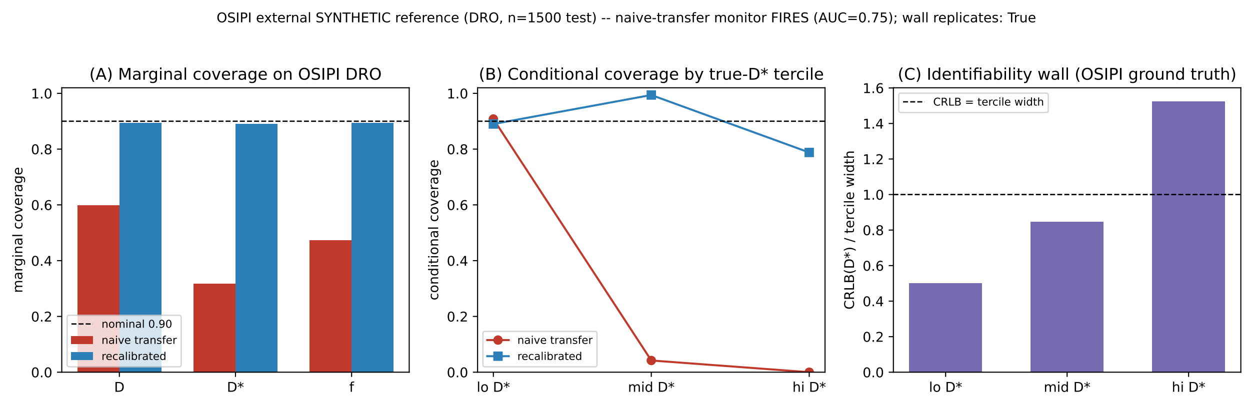

Figure 8: OSIPI external *synthetic* reference (DRO; not in vivo, no in-vivo coverage claim). (A) Marginal coverage vs nominal: naive transfer of our synthetic calibration breaks, recalibrating within the OSIPI distribution restores it. (B) Coverage conditional on the true D^* tercile: the high-D^* gap persists even after recalibration. (C) The identifiability wall on OSIPI ground truth – CRLB(D^*) exceeds the high-D^* tercile width. The naive-transfer monitor fires (AUC 0.74), so the break is flagged.

Supporting Information

The following figures support the main text and are provided as Supporting Information.

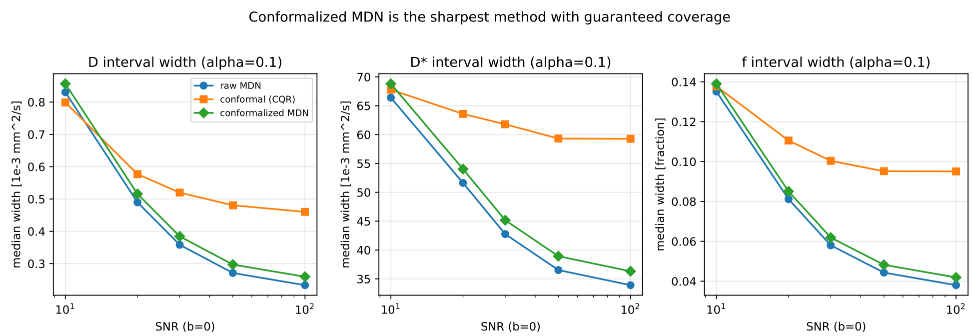

Figure S1: Median interval width vs SNR. Conformalized-MDN is the sharpest method with a guaranteed coverage level.

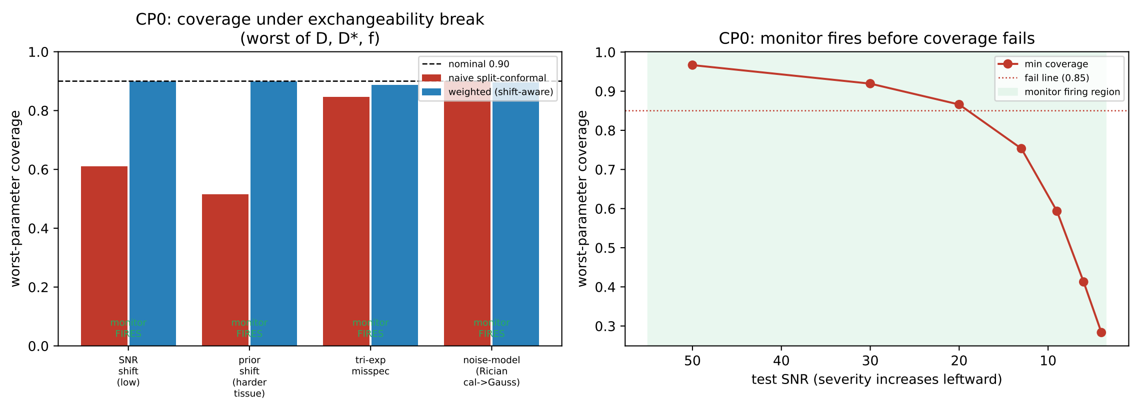

Figure S2: (Left) worst-parameter coverage under each exchangeability break, naive vs. weighted, with monitor status. (Right) SNR-severity sweep: the monitor fires before coverage crosses the failure line.

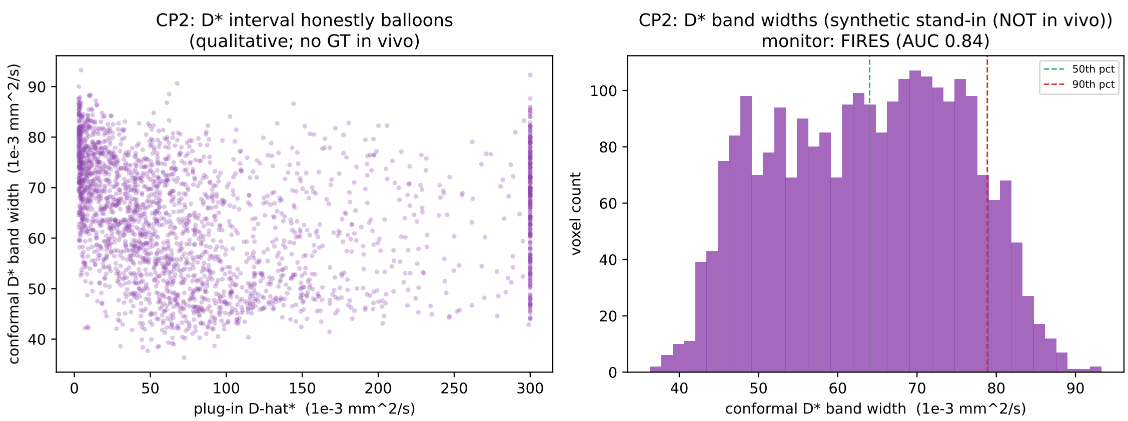

Figure S3: In-vivo qualitative demo (synthetic stand-in). (Left) the conformal D^* band balloons with the plug-in $\hat{D}^*$. (Right) the heavy-tailed D^* band-width distribution; the deployment monitor fires on the transfer.